CI/CD Observability in Containerized Environments

Project Preparation (PREPD)

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Context and Motivation

Why it matters

CI/CD services: operational infrastructure
Failures: block delivery and incident response
Diagnosis: required across multiple services

Current reality

Logs: scattered across systems
Correlation: weak across services
Decisions: incomplete signals

Problem Statement

Technical constraints

- **Non-privileged containers**: no host-level metrics
- **Telemetry fragmentation**: across CI/CD services
- **Mixed platforms**: Linux and Windows

Organizational constraints

- **No host privileges**: administrative access denied
- **Security policy**: privileged containers forbidden
- **Diagnosis**: slow and manual on-prem

Scope: non-privileged containerized CI/CD on mixed OS

Research Gap

What the literature assumes

- **Privileged access**: host or cloud-managed
- **Kernel mechanisms**: eBPF with elevated capabilities
- **Sidecar patterns**: Kubernetes contexts

Gap in practice

- **Non-privileged**: mixed OS deployments
- **CI/CD workloads**: limited empirical guidance
- **Ephemeral tasks**: post hoc correlation underexplored

Objectives and Research Question

Central question
How can an **integrated observability framework** be designed and evaluated for **privilege-restricted CI/CD** environments?

Objectives

- **Architecture**: container-first without host privileges
- **Prototype**: reproducible images + deployment templates
- **Evaluation**: controlled fault injection + operational metrics

Methodology Overview

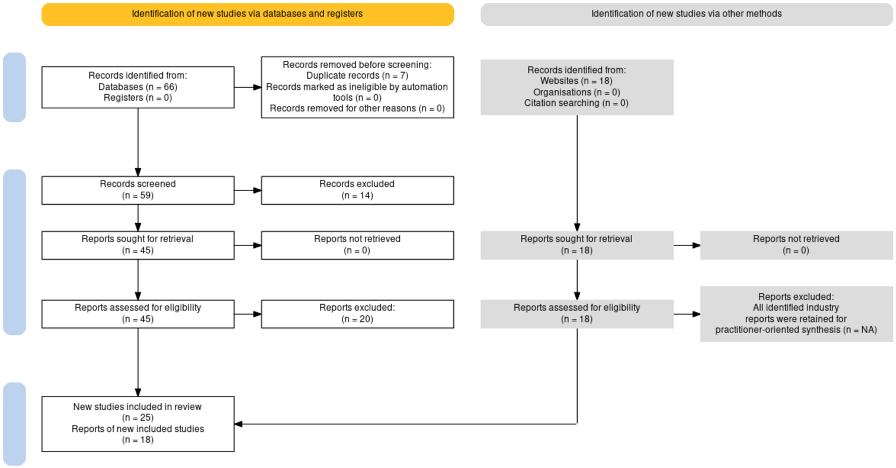
Design Science Research Approach:

Phase 1 **Problem analysis** + PRISMA literature review
Phase 2 **Technology assessment** and tool selection
Phase 3 **Architecture** specification + prototype implementation
Phase 4 **Evaluation campaign** with controlled fault injection
Phase 5 **Analysis**, thesis writing, defense preparation

Evaluation Strategy:

- **Quantitative**: MTTD, MTTR, resource overhead, throughput
- **Qualitative**: practitioner feedback via structured interviews

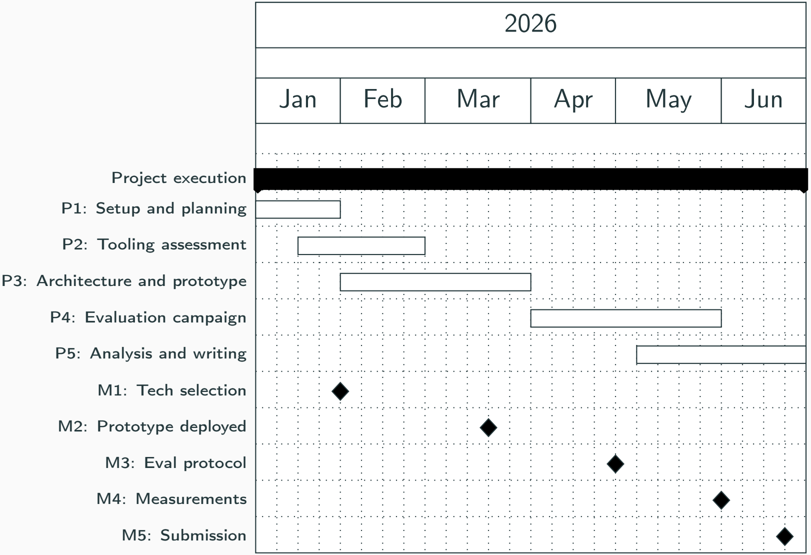
PRISMA Literature Review



PRISMA Summary

66 records identified (databases)
25 peer-reviewed studies included
18 industry sources retained

Project Plan and Timeline



Summary and Next Steps

Established

Problem framing: privilege-restricted CI/CD
PRISMA review: gaps identified
Methodology: execution plan defined

Next phase

Technology selection: constraint-driven
Prototype: architecture + implementation
Evaluation: controlled fault injection
Validation: practitioner feedback

Thank You

Questions?

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